Lecture Notes on

Special Relativity

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These notes summarize the twelfth Astrophysics Club meeting, where we introduced Einstein's Special Relativity. The central theme is that space and time are not absolute – they depend on the motion of the observer, with consequences that reshape our understanding of the universe.

Contents

1 What is Special Relativity?

Special Relativity is the framework Einstein published in 1905 describing how space and time behave at extreme speeds – close to the speed of light.

Key ideas:

- Space and time are not independent – they combine into spacetime.
- Measurements of length and time depend on the observer.
- The speed of light is the same for every observer.

At the time, many of Einstein's conclusions seemed physically impossible. Every prediction has since been confirmed by experiment – including gravitational lensing effects like the Einstein Cross.

Einstein's landmark paper was titled "On the Electrodynamics of Moving Bodies."

2 The Newtonian View of Space and Time

For roughly 2000 years, the geometry of space was considered intuitively obvious. This is the world of *Euclidean geometry*:

- Space is flat.
- Parallel lines never meet.
- The interior angles of a triangle sum to exactly 180°.
- Time flows the same for everyone, everywhere.

In this Newtonian picture, space and time are absolute – identical for every observer regardless of their motion.

3 Challenging Euclidean Assumptions

Italian mathematician Giovanni Saccheri (1667–1733) proved many results of non-Euclidean hyperbolic geometry, but ultimately dismissed them. He treated his own results as proof that such a geometry was impossible – he couldn't bring himself to accept what his math was telling him.

The lesson:

- Absolute space and time are assumptions, not derivations.
- These assumptions are what Einstein would eventually overturn.

4 The Einsteinian View

In Einstein's picture, space and time form a unified four-dimensional structure called *spacetime*. Mass and energy curve this structure, and objects follow the curvature.

Rough analogy: a heavy ball placed on a rubber sheet warps the fabric around it, and other objects roll toward it. This is only a caricature – real spacetime curvature is four-dimensional and includes time itself – but it captures the essential idea.

5 Frames of Reference

A **frame of reference** is a coordinate system from which measurements are made. Relativity is the study of how physical quantities measured in one frame relate to those measured in another.

An **inertial frame** is one that is neither accelerating nor rotating relative to another inertial frame.

Example: a spaceship moving at a constant speed v in a straight line away from Earth.

Special Relativity deals specifically with inertial frames.

6 Galilean Relativity

Long before Einstein, Galileo Galilei laid the groundwork for what we now call *Galilean relativity*.

Galileo's claim: Newton's second law

$$\sum \vec{F} = m\vec{a}$$

holds in all inertial frames.

Galileo further assumed that force, time, and mass are identical across frames:

$$F' = F, \quad t' = t, \quad m' = m.$$

Under Galilean relativity, velocities add in the ordinary way. If a train moves at 50 m/s relative to the ground and a ball is thrown forward at 10 m/s relative to the train, the ball moves at 60 m/s relative to the ground.

What is actually true:

- Only mass is the same across inertial frames.
- Force, acceleration, and time are *not* universal.
- These quantities depend on the observer's frame of reference.

Einstein would later show why.

7 The Problem with Galilean Relativity

In the 1860s, James Clerk Maxwell unified electricity and magnetism into a single elegant framework of four equations:

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}, \qquad \nabla \cdot \vec{B} = 0,$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \qquad \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}.$$

These equations predict electromagnetic waves that propagate at a fixed speed determined by fundamental constants:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3.0 \times 10^8 \text{ m/s}.$$

The problem: this speed is a fixed constant of the universe. It does not depend on the frame of reference. This directly contradicts Galilean relativity, which requires velocities to add and subtract between frames.

8 Einstein's Bold Interpretation

Most physicists of the era believed Maxwell's equations must be wrong, or that light required a medium – the "luminiferous aether" – to travel through.

Einstein took the bold step of accepting Maxwell's result as truth and rebuilding mechanics around it. The aether hypothesis was discarded; the constancy of c became a foundational fact of physics.

9 The Two Postulates of Special Relativity

9.1 Postulate 1 – Equivalence of Physical Laws

The laws of physics are identical for all observers moving at constant velocity. No experiment performed inside a closed system can determine whether it is at rest or in uniform motion.

9.2 Postulate 2 – Constancy of the Speed of Light

The speed of light in a vacuum is always

$$c \approx 3.0 \times 10^8 \text{ m/s}$$

for every observer, regardless of the motion of the light source or the observer measuring it.

9.3 Consequences

These two postulates, taken together, force us to abandon absolute time and absolute space. Every strange consequence of Special Relativity follows from them.

10 The Lorentz Transformations

The Lorentz transformations are the mathematical heart of Special Relativity. They convert positions and times measured in one inertial frame into those measured in another frame moving at speed v along the x-direction:

$$ct' = \gamma\left(ct - \beta x\right), \qquad x' = \gamma\left(x - \beta ct\right).$$

Key quantities:

$$\beta = \frac{v}{c}, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-(v/c)^2}}.$$

Here γ is the **Lorentz factor** – the single quantity that controls every relativistic effect.

The Lorentz transformations were first derived by Hendrik Lorentz, but Einstein was the one who gave them their correct physical interpretation.

11 The Lorentz Factor

The Lorentz factor is

$$\gamma = \frac{1}{\sqrt{1-(v/c)^2}}.$$

Behavior:

- At $v = 0$, $\gamma = 1$ – no relativistic effects.
- At $v = 0.5c$, $\gamma \approx 1.15$ – small correction.
- At $v = 0.99c$, $\gamma \approx 7.09$ – dramatic.
- As $v \to c$, $\gamma \to \infty$.

This last point is why nothing with mass can reach the speed of light – it would require infinite energy.

12 Length Contraction and Time Dilation

The Lorentz transformations lead to two famous consequences.

12.1 Length Contraction

An object moving relative to you appears *shorter* in the direction of motion:

$$L = \frac{L_0}{\gamma},$$

where L_0 is the length measured in the object's rest frame (the *proper length*). The faster it moves, the more its measured length contracts.

12.2 Time Dilation

Moving clocks run slow relative to your frame:

$$t = \gamma\tau,$$

where τ is the time measured in the moving frame (the *proper time*). As speed increases, the time between events grows longer in your frame.

Both effects are real, symmetric, and confirmed by experiment.

13 Recovering Everyday Physics

For everyday speeds, $v \ll c$, so

$$\gamma = \frac{1}{\sqrt{1-(v/c)^2}} \approx 1.$$

In this limit, the Lorentz transformations reduce to the ordinary Galilean ones, and Newton's second law holds in every inertial frame.

Special Relativity does not replace Newtonian mechanics – it *contains* Newtonian mechanics as a low-speed limit.

14 The Twin Paradox

The Twin Paradox is the most famous thought experiment arising directly from Special Relativity.

14.1 The Setup

Two twins are born on Earth on the same day. Immediately after birth, one twin departs on a spaceship traveling at $v = 0.9c$. The traveling twin experiences 20 years of life aboard the ship. Upon returning to Earth, the traveling twin finds their sibling has aged 46 years.

14.2 The Math

Time dilation gives $t_{\text{earth}} = \gamma\, t_{\text{spaceship}}$, with

$$\gamma = \frac{1}{\sqrt{1-(0.9)^2}} \approx 2.29.$$

Therefore

$$t_{\text{earth}} = 2.29 \times 20 \approx 46 \text{ years}.$$

14.3 Why It Matters

Time dilation has been confirmed experimentally using atomic clocks on aircraft and GPS satellites. Many people assume relativistic effects can be ignored at everyday speeds, but the corrections are essential for day-to-day systems like GPS navigation.

15 Mass-Energy Equivalence

Perhaps the most famous consequence of Special Relativity is that mass and energy are two forms of the same thing:

$$E = mc^2.$$

The conversion factor $c^2 \approx 9 \times 10^{16}\ \text{m}^2/\text{s}^2$ is enormous. A small amount of mass corresponds to a huge amount of energy.

16 Why Stars Shine for Billions of Years

Inside the cores of stars like our Sun, hydrogen nuclei fuse to form helium. The helium nucleus has slightly less mass than the protons that formed it. That missing mass – the **mass defect** – is converted directly into energy:

$$E = \Delta m\, c^2.$$

Even a tiny mass defect releases enormous energy, which is why stars can sustain nuclear fusion for billions of years.

Applications:

- Stellar fusion powers every star in the universe.
- Nuclear reactors convert small mass defects into usable electricity.
- Nuclear weapons convert tiny fractions of ^{235}U's mass into destructive energy.

17 Big Picture

Special Relativity dismantles Newton's assumptions about space and time and replaces them with a framework consistent with electromagnetism.

Key takeaways:

- The speed of light is a fundamental constant of nature.
- Space and time are relative to the observer.
- Mass and energy are equivalent.
- Newtonian mechanics is recovered at low speeds.

The next step, *General Relativity*, extends these ideas to include gravity and accelerating frames – and gives us the modern picture of curved spacetime that underlies black holes and cosmology.